

Low-Level Design (LLD) Document Template

Reference Template for Document Validation System

1. Introduction

1.1 Purpose

Describe the purpose of this Low-Level Design (LLD) document. Mention the system name and explain what detailed design this document is intended to define.

1.2 Scope

Define the scope of the detailed design specifications covered in this document.

1.3 Definitions, Acronyms, and Abbreviations

List important terms, acronyms, and abbreviations used in the document.

Term	Description
LLD	Low-Level Design
Module	A self-contained component of the system
API	Application Programming Interface

1.4 References

List related documents such as the HLD, SRS, FRD, and coding standards.

2. Module Design

2.1 Module Overview

Describe the main modules of the system, their responsibilities, and their dependencies.

Module	Responsibility	Dependencies
User Manager	User registration and authentication	Database
Document Processor	Parse and extract content from files	File I/O
Validation Engine	Compare documents against templates	AI service
Report Generator	Generate validation reports	Document Processor
Template Manager	Manage document templates	Database

2.2 Module Details

Provide a detailed description for each module, including its methods, parameters, return values, and error handling.

3. Class and Interface Design

3.1 Class Diagram

Insert UML class diagrams showing class structures and their relationships.

3.2 Key Classes

Describe the main classes used in the system.

- User - represents a system user with credentials and profile
- Document - represents an uploaded document with metadata
- Template - represents a document template for validation
- ValidationResult - stores validation scores and results

4. Function and Method Specifications

4.1 API Endpoints

Describe the main API endpoints, their methods, and parameters.

Endpoint	Method	Description
/api/users/register	POST	Register a new user
/api/users/login	POST	Authenticate a user
/api/documents/upload	POST	Upload a document
/api/validation/validate	POST	Validate a document
/api/templates/list	GET	Retrieve all templates

4.2 Method Details

Provide detailed specifications for critical methods, including algorithm descriptions and error handling.

5. Database Design

5.1 Database Schema

Provide the detailed schema with tables, columns, data types, and constraints.

5.2 Entity-Relationship Diagram

Insert the ER diagram showing all tables and their relationships.

5.3 Indexing Strategy

Describe the indexing approach used to optimize queries.

- Index on user email for login queries
- Index on document owner for user document retrieval
- Index on document ID for validation result queries

6. Algorithm Design

6.1 Key Algorithms

Describe the main algorithms used in the system. For example, the document validation algorithm:

- Extract text from the uploaded document
- Identify document sections
- Compare against the reference template structure
- Calculate a structure score based on missing and extra sections
- Analyse content quality
- Generate the final validation result

6.2 Data Structures

Describe the key data structures used to represent documents and templates.

7. Error Handling and Logging

7.1 Exception Handling

Define how errors and exceptions are handled across the system.

- File format errors for unsupported file types
- Validation errors during document processing
- Database errors during data operations

7.2 Logging Strategy

- Log all API requests and responses
- Log validation process steps and results
- Log errors with meaningful messages

8. Coding Standards

8.1 Naming Conventions

Define the naming conventions used for classes, methods, and constants.

8.2 Code Organization

- Follow the single responsibility principle
- Avoid code duplication
- Use design patterns where appropriate
- Keep code organized for easy maintenance

9. Validation Rules

9.1 Required Sections

The uploaded LLD document should include the following main sections:

- Introduction
- Module Design
- Class and Interface Design
- Function and Method Specifications
- Database Design
- Algorithm Design
- Error Handling and Logging
- Coding Standards
- Validation Rules
- Appendices

9.2 Validation Criteria

The document validation system should check whether required sections are available, whether section names are similar to the LLD template, whether sections follow the correct order, and whether the document belongs to the correct document type.

9.3 Scoring Criteria

Score Range	Validation Result
75-100	Accept
50-74	Needs Improvement
0-49	Reject

10. Appendices

10.1 Appendix A: Code Examples

Include sample code snippets demonstrating key implementations.

10.2 Appendix B: Additional Information

Include any extra information related to the detailed design.